

Time: 80 mins

- c. From the circuit of Figure 1 determine the currents and voltages of each branch. Considering R_3 as load find the load line. [CO1] 5

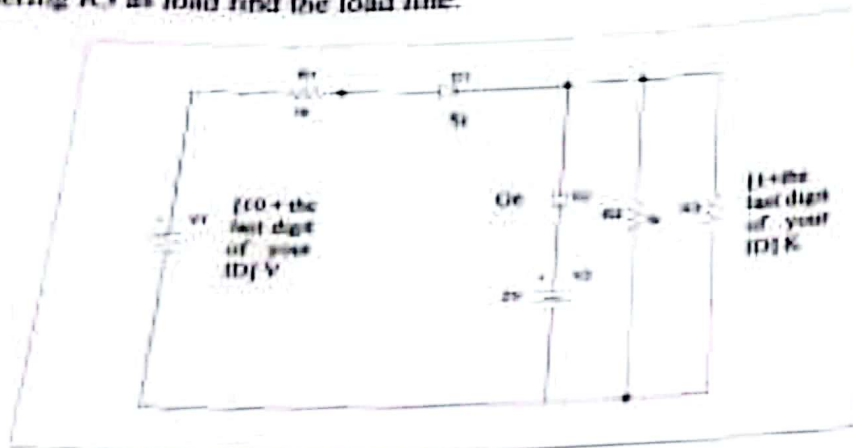


Figure 1

2. Consider the circuit of Figure 2. A source (V_s) with a peak amplitude of 24V is employed to charge a 12V dc-battery. (a) Find the fraction of each cycle during which the diode conducts. (b) Find the peak value of the diode current and maximum reverse-bias voltage that appears across the diode. [CO1] 5

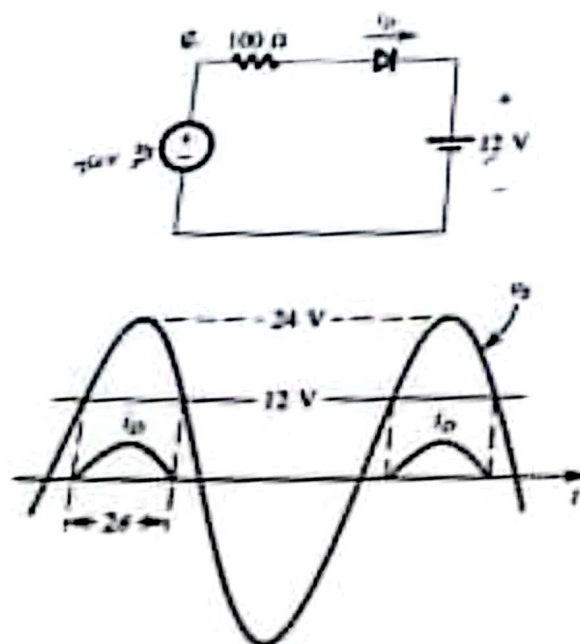


Figure 2

3. Draw a full wave bridge rectifier circuit with [4+the last digit of your ID] K Ω load resistance and sketch the output waveforms when the input is [10+the last digit of your ID] V sinusoidal signals. Also, find the dc value of the full wave bridge rectifier circuit. [CO2] 5
4. For the network given in Figure 3, the input signal V_2 is given as a sinusoid with [40+the last digit of your ID] V peak to peak. Analyze the circuit and sketch the waveforms of output with proper labeling. [CO2] 5

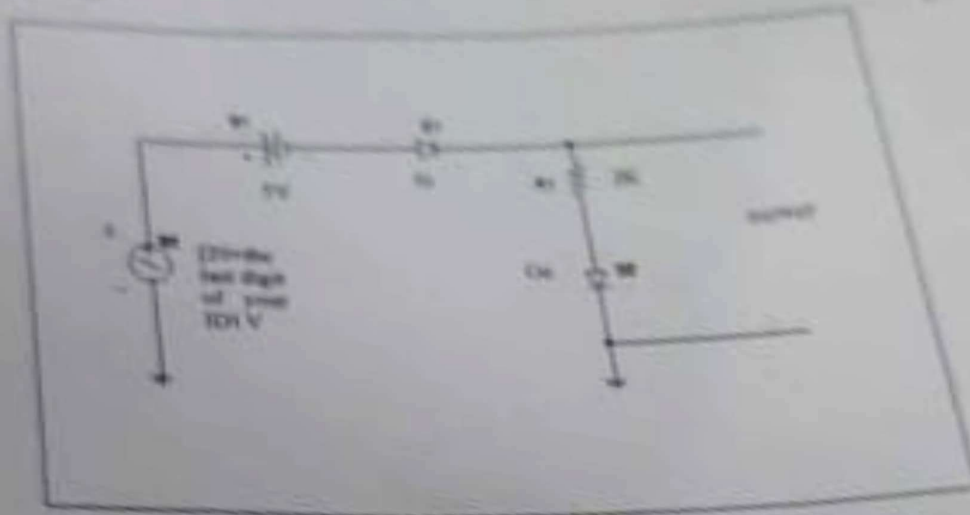


Figure 3

5. Find the voltage and current of the R_1 resistor, R_2 resistor, Zener diode, and the Zener power for the circuit of Figure 4, if $R_2 = 4K\Omega$. [CO2] 5

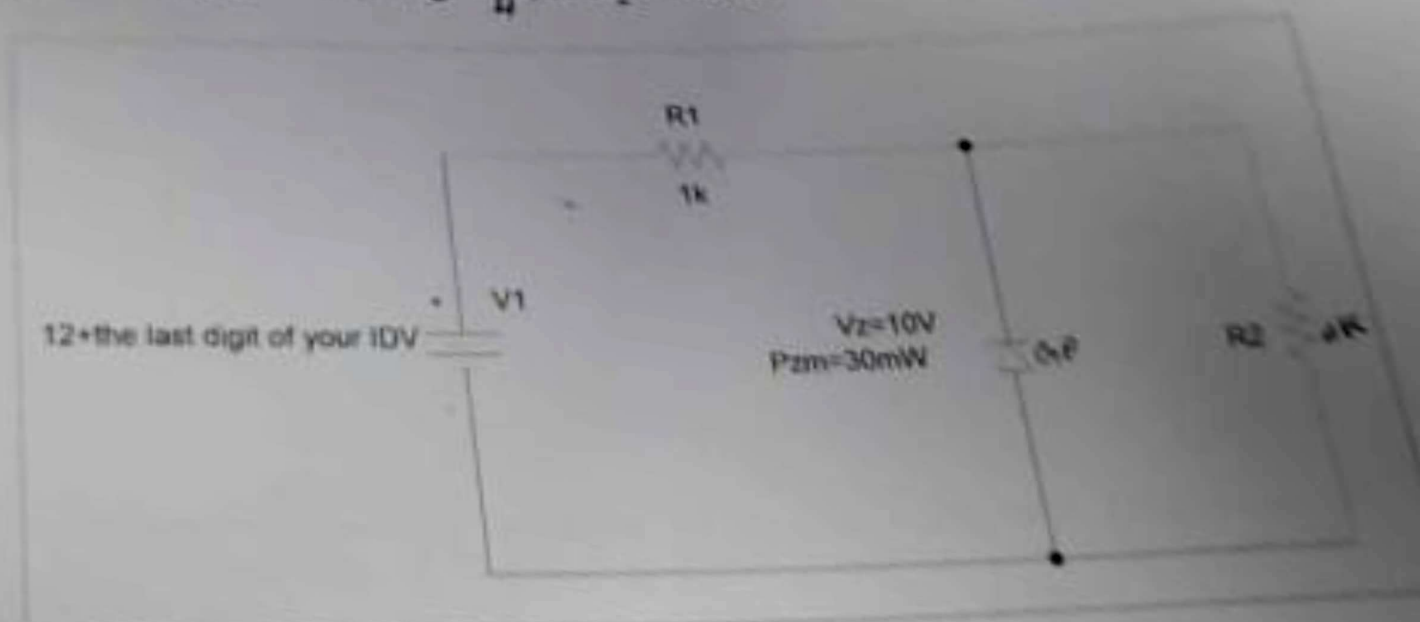
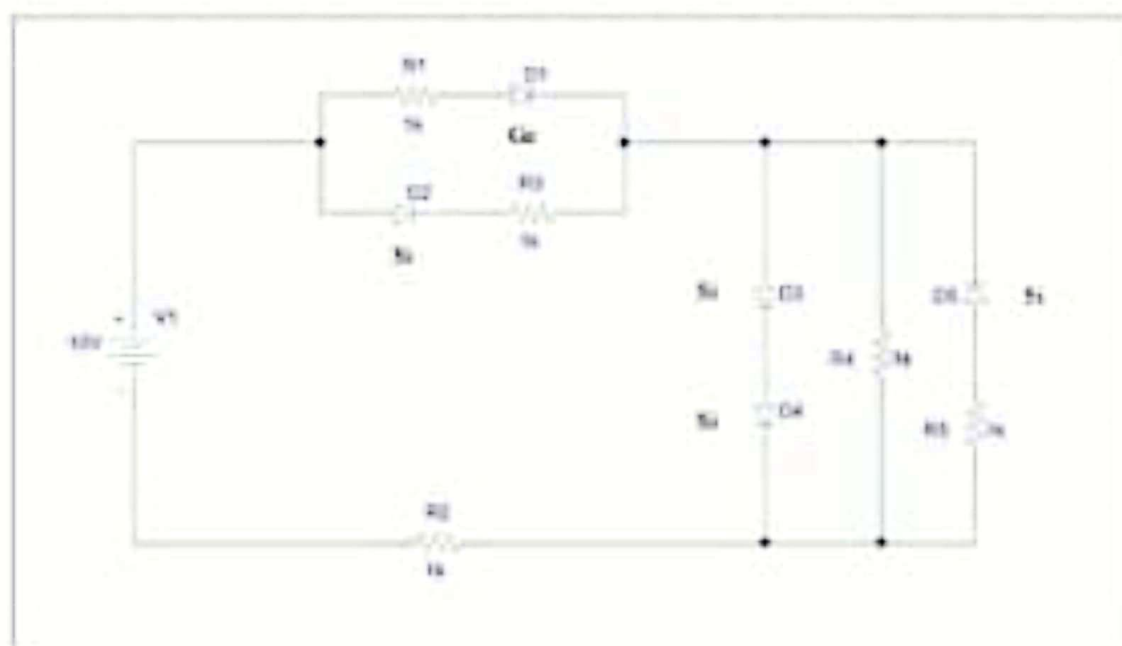


Figure 4

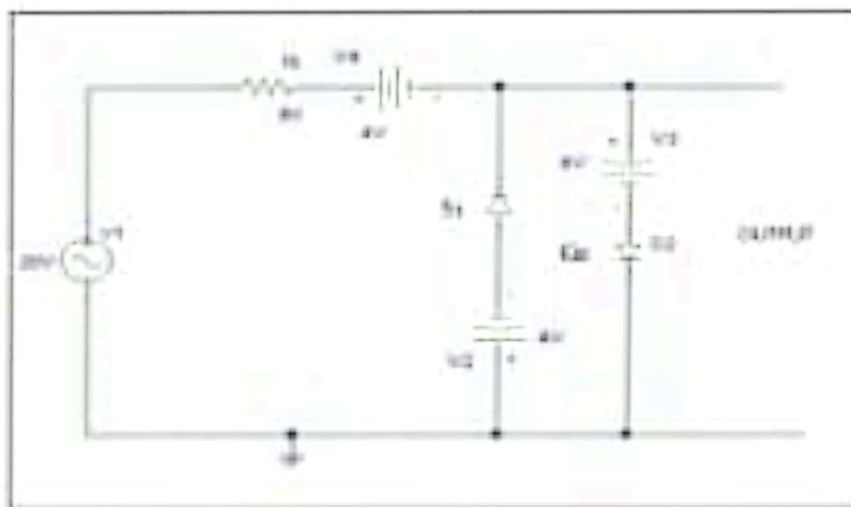
- Describe with an appropriate diagram in your own words the conditions established by forward and reverse bias conditions on a p-n junction diode and how the resulting current is affected. [CO1]5
 - Find the resistivity of (a) intrinsic silicon and (b) p-type silicon with $N_A = 10^{16}/\text{cm}^3$. Use $n_i = 1.5 \times 10^{10}/\text{cm}^3$, and assume that for intrinsic silicon $\mu_n = 1350 \text{ cm}^2/\text{V}\cdot\text{s}$ and $\mu_p = 480 \text{ cm}^2/\text{V}\cdot\text{s}$, and for the doped silicon $\mu_n = 1110 \text{ cm}^2/\text{V}\cdot\text{s}$ and $\mu_p = 400 \text{ cm}^2/\text{V}\cdot\text{s}$. (Note that doping results in reduced carrier mobilities.) [CO1]5
- Determine a) DC load line b) Voltage across the resistance c) Instead of the DC source use a 10V AC source and find the AC load line. (Consider an approximate diode characteristics curve.) [CO2]5



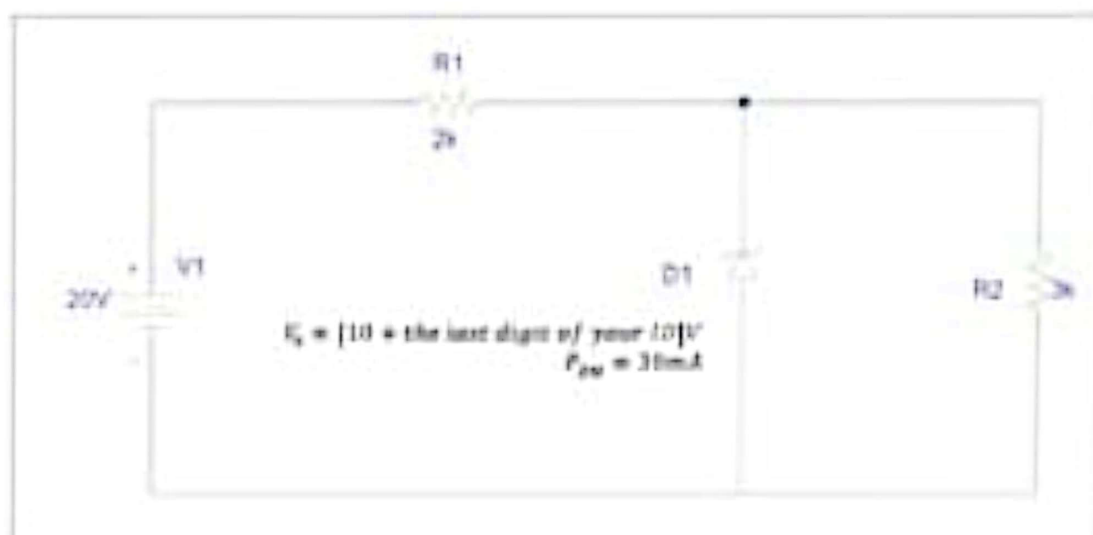
- Determine the currents through R_1 , R_3 , and R_4 resistances respectively. [CO2]10



4. Draw a full-wave rectifier circuit with $4\text{ k}\Omega$ load resistance and sketch the output waveforms when the input is 10V sinusoidal signals. Also, find the dc values of the circuits. [CO2]5
5. Draw the output waveforms for the following circuits. [CO2]10

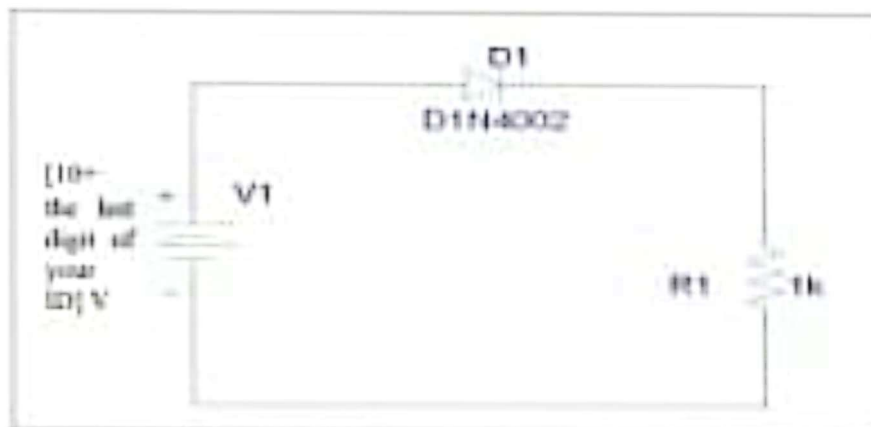


6. Determine V_2 , I_2 , I_R , I_{R1} , and P_R from the following figure. [CO2]10

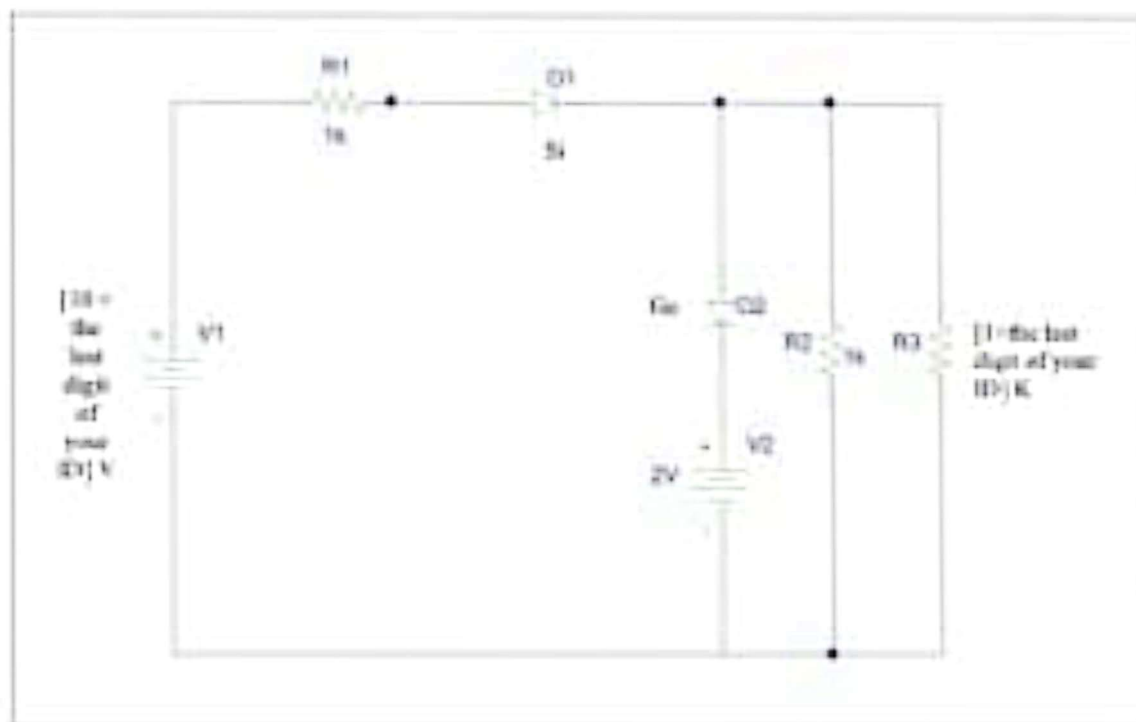


forward and reverse bias conditions on a p-n junction diode and how the resulting current is affected. 5

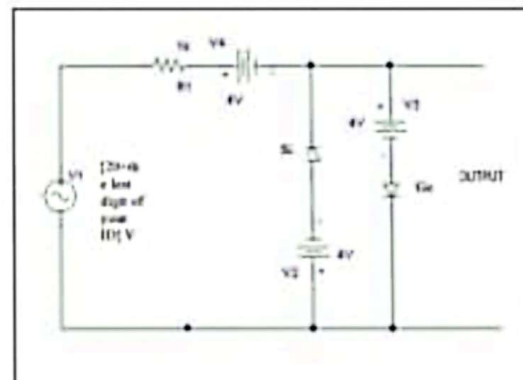
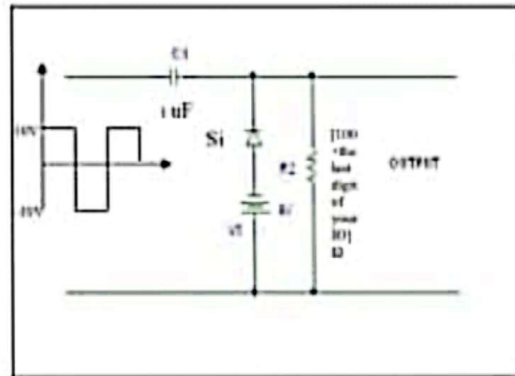
2. Determine a) DC load line b) Voltage across the resistance c) Instead of the DC source use a 10V AC source and find the AC load line. (Consider an approximate diode characteristics curve.) 1+2+2=5



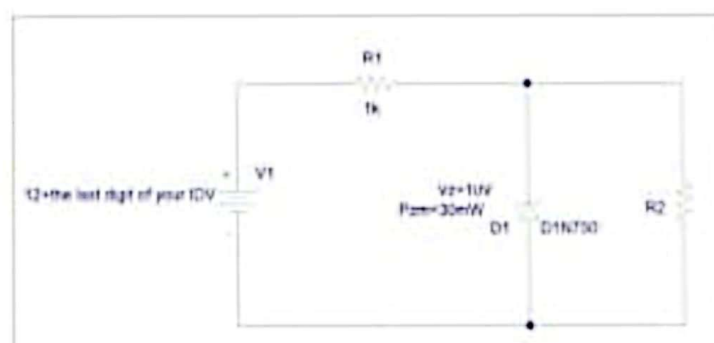
3. From the following circuit determine the currents and voltages of each branch. 5+5=10



4. Draw a half wave and full wave rectifier circuit with $[4 + \text{the last digit of your ID}] \text{ K}\Omega$ load resistance and sketch the output waveforms when the input is $[10 + \text{the last digit of your ID}] \text{ V}$ sinusoidal signals. Also find the dc values for both the circuits. $4 + 4 + 2 = 10$
5. Draw the output wave forms for the following circuits. $5 + 5 = 10$



6. Determine V_L , I_L , I_D and P_L for the following figure if $R_2 = 1.2 \text{ K}\Omega$. Repeat the problem with $R_2 = [3 + \text{the last digit of your ID}] \text{ K}\Omega$. $2 + 2 + 2 + 2 = 10$

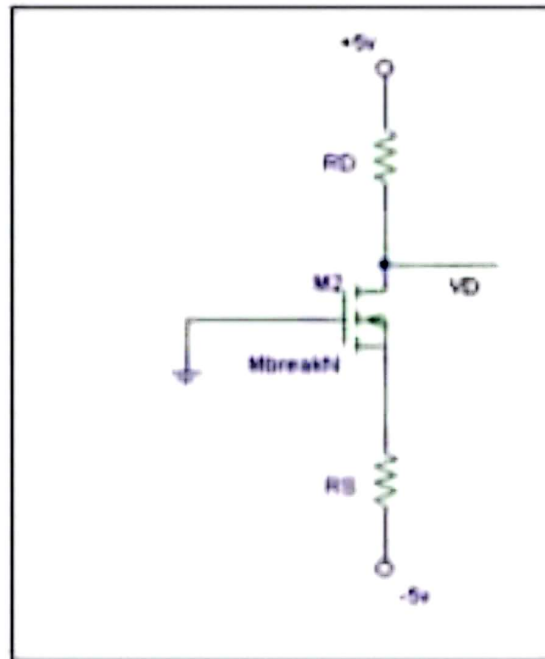


1. a. Sketch an n-channel enhancement type MOSFET with proper biasing applied ($V_{DS} > 0V$, $V_{GS} > V_t$) and indicate the channel, the direction of electron flow, and the resulting electron flow.

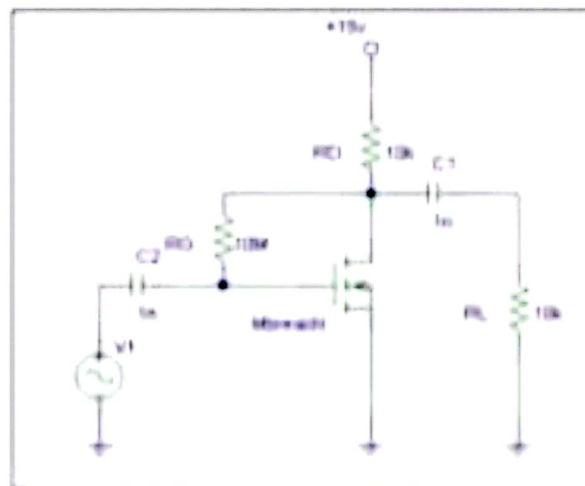
b. In your own words, briefly describe the basic operation of an enhancement type MOSFET.

[CO3] 5+5=10

2. Find V_{GS} , R_S and R_D for the following circuit so that the transistor operates at $I_D = 0.4\text{mA}$ and $V_D = +1V$. The NMOS transistor has $V_t = 2V$, $\mu C_{OX} = (20 + \text{the last digit of your ID})\mu\text{A}/V^2$, $L = 10\mu\text{m}$ and $W = 400\mu\text{m}$. Neglect the channel length modulation effect (i.e assume that $\lambda = 0$). [CO3] 10

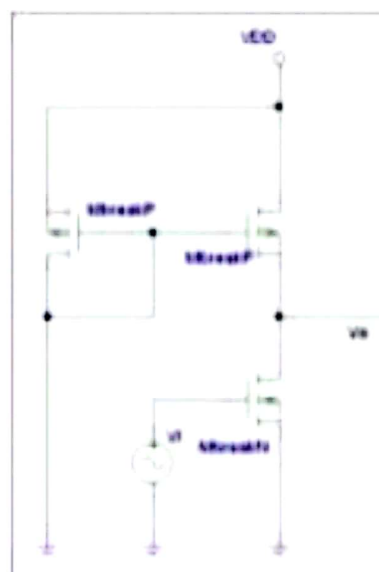


3. The following circuit shows a discrete enhancement MOSFET amplifier in which the input signal v_i is coupled to the gate via a large capacitor, and the output signal at the drain is coupled to the load resistance R_L via another large capacitor. Draw the small-signal equivalent circuit, determine voltage gain and its input resistance. The transistor has $V_t = 1.5\text{ V}$, and $V_A = (50 + \text{the last digit of your ID})\text{ V}$, and $k_n'(W/L) = 0.25\text{ mA/V}^2$. Assume, the coupling capacitor to be sufficiently large to act as a short circuit at the signal frequencies of interest. [CO3] 3+4+3=10



4. a. Draw the circuit diagram of a common source, common gate, and common drain circuit.
b. Draw the transfer characteristics, the equivalent circuit of the following circuit.

[CO3] 5+5=10





Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Assignment, Summer 2023 Semester

Course: Electronic Circuits
(CSE251|Section:3|ICE213|Section:3|ICE251|Section:6)
Full Marks: 15

1. From the circuit of **Figure 1** determine the currents and voltages of each branch. Considering R_3 as load find the load line.

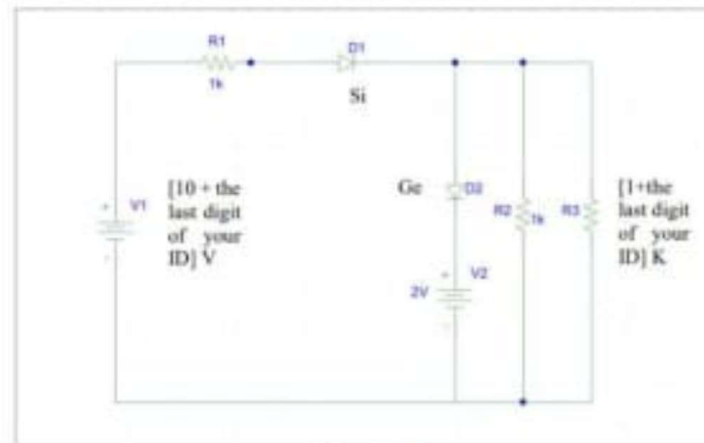


Figure 1

2. For the network given in **Figure 2**, the input signal V_2 is given as a sinusoid with $[40 + \text{the last digit of your ID}] \text{V}$ peak to peak. Analyze the circuit and sketch the waveform of output with proper labeling.

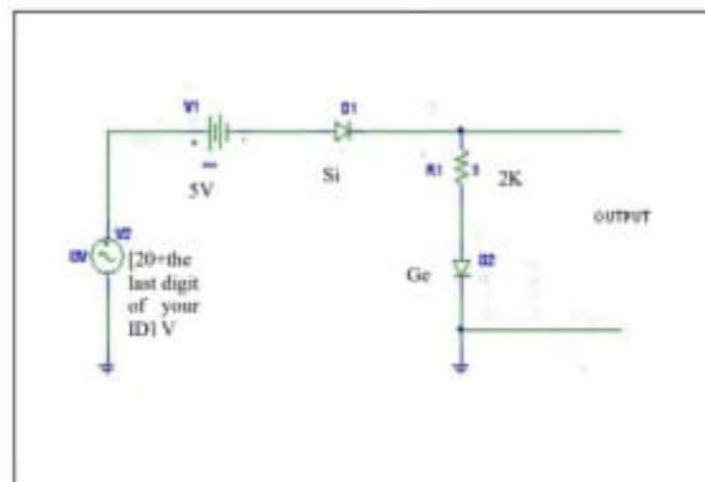


Figure 2

3. Determine the levels of I_{CQ} and V_{CEQ} for the voltage divider configuration of **Figure 3**. Show the dc load line analysis also. Analyze the following circuit for a small signal model and determine the small-signal voltage gain A_v , and small signal current gain A_i .

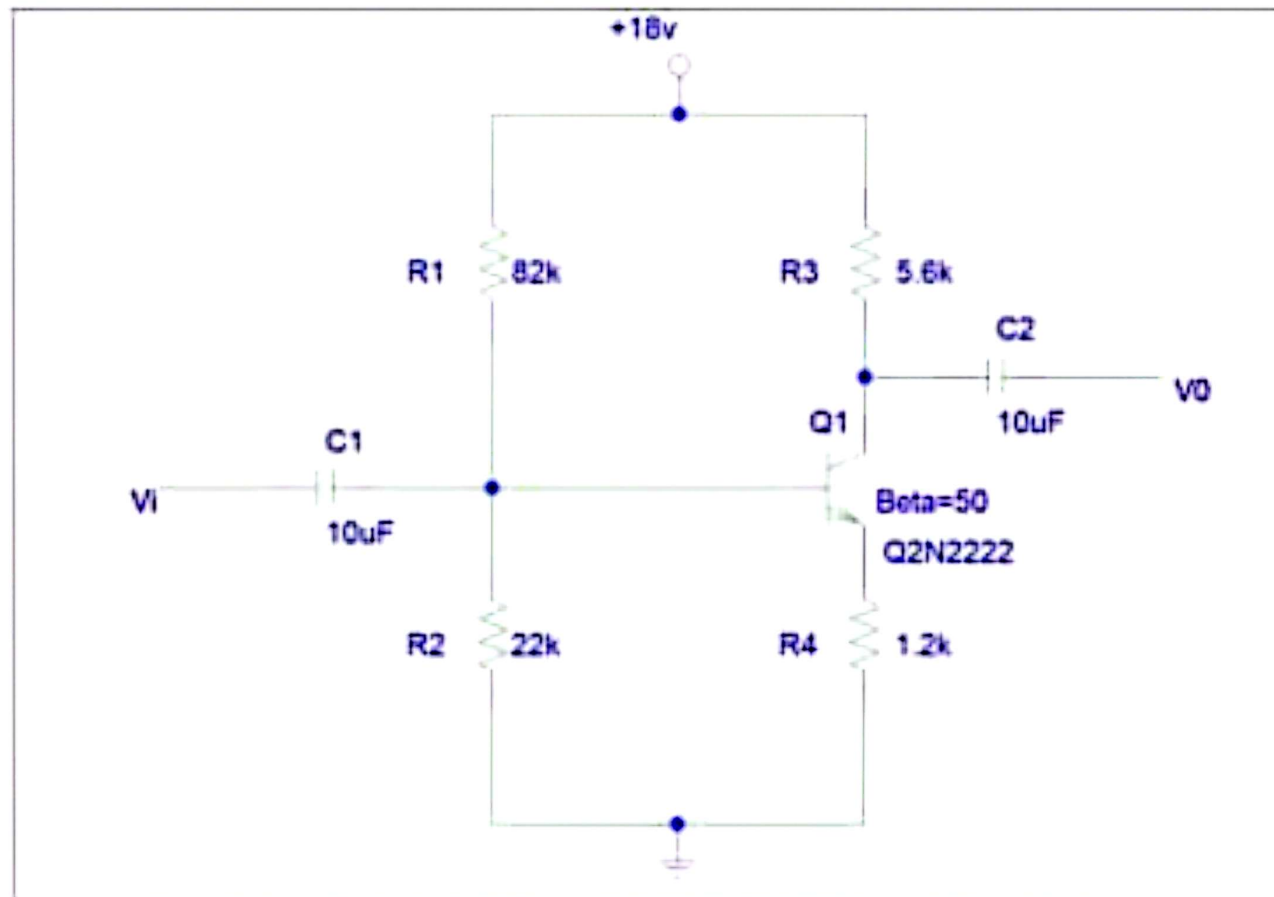


Figure 3